

LEDGER · 2026-09-01 · MATERIALS SCIENCE

Long-Afterglow Strontium Aluminate Phosphors via Sol-Gel Auto-Combustion

REJECTED

This concept did **not** clear the framework.
The reasoning is published in full below.
What the assessment found

Incumbent benchmark	Commercial bulk solid-state SrAl2O4:Eu,Dy
Entry application	Photoluminescent safety signage and emergency egress pathmarking
Measured advantage	<0.5x afterglow duration (<5 hours vs 10-24 hours for incumbent)
Time to first revenue	4 months
Gross margin at parity	77.5%
Startup CapEx	\$118,500
Payback from first sale	3.3 months
Capital productivity	6.38x

Regulatory risk

Moderate — qualification costs reported (\$45,000 estimated)

Why it failed

- Voided by: Negative control / sub-2x, Form-factor / handling regression
- Below the 2x step-change bar

Assessment

The report explicitly concedes functional failure because nanoscale auto-combustion SrAl_2O_4 suffers from severe surface quenching, reducing afterglow duration to $<0.5x$ of the solid-state incumbent. In addition, the nanoscale form factor creates severe handling/inhalation hazards and accelerates moisture degradation.

Also flagged

- Price parity asserted, not sourced

A rejection is not a claim that the science is wrong. It means the venture case did not survive: the comparator was not what the buyer actually buys, the comparison was not like-for-like, the advantage fell short of a step change, or the capital and time to first revenue were prohibitive.

Assessed 2026-09-01 against seven gates plus the voiding sub-tests. [How the method works](#) · [Browse all concepts](#)

The Absence Audit

Method

Ledger

Free studies

Track record

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